# AI and Computer Vision based Grocery Shop Automation

## Abstract

This project proposes a computer vision and AI–based grocery shopping automation system that reduces manual scanning, sorting, and billing effort in small to medium grocery shops. A customer first submits a grocery list by showing it to a camera; the system captures the image and converts it to text using an AI-based image-to-text method to extract item names and quantities, then uploads the structured order to a central server and database. The shopkeeper views incoming orders on a laptop or mobile device, and the system also flags item shortages from inventory records so the shopkeeper can respond quickly. Collected items are placed sequentially on a conveyor belt where a fixed camera performs real-time video processing and object detection using YOLOv8 to recognize and count products. To support multiple customers at the same time, the detection results are mapped to each customer's scanned list, enabling per-customer tracking without requiring the shopkeeper to pre-sort items. Finally, the system determines which detected items belong to which customer basket and confirms completion through item-wise counting. The initial prototype will be designed to handle up to three customers concurrently, demonstrating reliable list extraction, shortage notification, multi-order tracking, and automated item-to-basket allocation.

## Budget

| **Item** | **Price (BDT)** |
|---|---|
| NEMA 23 stepper motor + TB6600 driver (x1) | 4,650 |
| MG995 servo motor (x3) | 1,080 |
| ESP32 Type-C dev board (x2) | 1,100 |
| Breadboard 830 tie-points (x2) | 260 |
| Jumper wires M2M set, 40pcs (x2) | 200 |
| Jumper wires M2F set, 40pcs (x2) | 200 |
| Jumper wires F2F set, 40pcs (x2) | 200 |
| NEMA 23 motor frame / L-bracket (x1) | 699 |
| Angle bar 3 ft (x2) | 420 |
| Belt (GT2 timing belt 6mm, 1m) (x1) | 220 |
| PVC pipe 3/4" with socket, 10 ft (x2) | 470 |
| M6 bolt/screw (M6 cup head, per piece) (x2) | 40 |
| Raspberry Pi 5 Model B (8GB) (x1) | 14,900 |
| Raspberry Pi Camera Module 3 (x1) | 6,920 |
| **Total Budget** | 31,359 |